

Data Sources and Methods

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Weeks 4 Progress Update:

(9/8/2026 - 9/14/2026)

Focus

Reviewed literature for environmental factors to include in the species distribution model.

Candidate Variables

The following variables are currently selected based off representation in literature, percieved data availability, or in an effort to maintain cross comparison compatibility with the HSI derived in Geselbracht et al. (2024).

Table 1: Preliminary Environmental Variables Considered for the Eastern Oyster SDM

Variable	Reason	Reference	Data Source
Salinity	Widely available, large driver of ecosystem suitability	(lePeyre2021? ; Geselbracht et al. 2024; Pollack et al. 2012;Barnes et al. 2007)	Water Quality Portal (DeCicco et al. 2026)
Water temperature	Widely available, large driver of ecosystem suitability	(lePeyre2021? ; Geselbracht et al. 2024; Pollack et al. 2012;Barnes et al. 2007)	Water Quality Portal (DeCicco et al. 2026)

Table 1: Preliminary Environmental Variables Considered for the Eastern Oyster SDM

Variable	Reason	Reference	Data Source
Substrate / bottom type	suitable substrate required for recruitment	(lePeyre2021? ; Geselbracht et al. 2024; Barnes et al. 2007)	TBD
Water depth / bathymetry	Depth is a key driver in oyster habitat extent	(lePeyre2021? ; Geselbracht et al. 2024; Pollack et al. 2012; Barnes et al. 2007)	National Centers for Environmental Information (National Centers for Environmental Information 2018)
Dissolved oxygen	Low DO can cause oyster stress and mortality	(lePeyre2021? ; Geselbracht et al. 2024; Pollack et al. 2012)	Water Quality Portal (DeCicco et al. 2026)
Chlorophyll- a	Common proxy for food resources supporting oyster growth	(lePeyre2021? ; Geselbracht et al. 2024)	Water Quality Portal (DeCicco et al. 2026)
Oyster bed locations	important for SDM modeling requiring species presence data	(lePeyre2021? ; Geselbracht et al. 2024)	Florida Fish and Wildlife Conservation Commission (Florida Fish and Wildlife Conservation Commission 2015)

Week 5 Progress Update:

(9/15/2026 - 9/21/2026)

Focus

Review literature relating to spatial modeling methods to employ in the interpolation of geographic variables to create a continuous surface grids of predictor variables, and review the MaxEnt species distribution modeling logic.

Methods

Spatial interpolation of environmental variables

MaxEnt Species Distribution Modeling

*****Maximum entropy modeling of species geographic distributions***** Phillips, Anderson, and Schapire (2006)

- Background

The authors introduce Maxent as a method for species distribution modeling when only presence data is available. The authors cite that for many species having both presence and absence data is very unlikely. Maxent uses environmental characteristics of known locations and is able to apply them to unknown locations to predict probability of species occurrence.

Niche-based models can predict the geographic extent, but will often overestimate the reality of the species distribution. It is recognized that because an environment might fall within the niche of a given species, doesn't necessarily mean the species are actually established there due to a multitude of geographic, biotic, or anthropogenic barriers. Additionally, there are many assumptions made in such modeling techniques, so the authors express that much care must be taken when choosing which environmental factors to use. With so many assumptions, the authors cite many sources of errors that can be introduced if care is not taken such as using improper temporal or spatial resolutions and scales. Further, the authors state significant errors can also arise from improper data manipulation, poorly optimized environmental models, and interpolation of lower-resolution data.

We must also acknowledge that there are problems with presence-only data as a whole that must be considered when designing the model. The authors mention things such as geographic sampling bias, spatial autocorrelation, poor sampling design (e.g. small sample sizes and erroneous sample collection/reporting), and missing or important environmental predictors to name a few.

- Maxent

The authors describe the basics of maximum entropy modeling where the goal is to estimate an unknown probability distribution while making as few assumptions as possible. As hinted in the name, the model seeks the distribution that has the maximum entropy (i.e. the most uniform distribution), while maintaining the influence of what is known from the presence data. In species modeling, the model is made up of 3 major parts: 1) the study area pixel grid represents all possible locations, 2) known presence observations provide sample locations, and 3) the environmental predictor variables act as features that describe the conditions at the species occurrence locations.

The authors highlight a number of advantages and disadvantages associated with Maxent modeling.

Maxent advantages

- Only requires presence data;
- Can use both continuous and categorical data, and incorporate interactions between variables;
- Efficient deterministic algorithms have been developed which are guaranteed to converge on optimal probability distribution;
- The Maxent probability distribution has a well described mathematical definition;
- Over-fitting can be avoided by using regularization techniques;
- Provides flexibility for adding thresholds to continuous predictions;
- Can work on relatively limited data and be applied to all levels of scale.

Maxent disadvantages

- At the time of the paper, the Maxent method was still new;
- model guidance was still under development;
- uncertainty estimation was not as well developed when compared to other statistical methods;
- overestimation of environmental variables is possible due to the use of an exponential model for probabilities;
- specialized software requirements

References

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- DeCicco, Laura, Robert Hirsch, David Lorenz, Jordan Read, Jordan Walker, Lindsay Platt, David Watkins, et al. 2026. *dataRetrieval: R Packages for Discovering and Retrieving Water Data Available from u.s. Federal Hydrologic Web Services* (version 2.7.25). Reston, VA: U.S. Geological Survey; U.S. Geological Survey. <https://doi.org/10.5066/P9X4L3GE>.
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